

CRM Platform

Consolidated Solution Document

Multi-Tenant SaaS · full-platform overview · June 2026

1. Executive summary

A multi-tenant SaaS CRM and customer-operations platform. Each customer ("tenant") gets an isolated workspace spanning core CRM, sales & invoicing, support ticketing, voice (incl. an AI voice bot), shared email, team messaging, integrations and analytics. A provider-level Super Admin console manages tenants, plans, licensed modules, platform billing and access. Licensing is enforced through a layered access-control model so each tenant only reaches what was sold to them.

2. Architecture

Monorepo (npm workspaces): packages/{api, core, frontend, mobile, shared} and a pluggable modules/* directory loaded by a module registry.

Layer	Technology
API	Fastify (JWT, Helmet, CORS, rate-limit, multipart, Swagger/OpenAPI), Zod validation, GraphQL via Mercurius
Data	PostgreSQL with Row-Level Security for tenant isolation; Redis for caching, tenant cache & rate-limit counters
Frontend	React + React Router, TanStack Query/Table, Zustand, Tailwind, Recharts, react-hook-form + Zod, dnd-kit
Realtime / Voice	LiveKit (server + client SDK) for calls and recordings
Documents	docxtemplater + PizZip for invoice template rendering; nodemailer for outbound email
Mobile	Dedicated mobile package (shared types via @crm/shared)

3. Multi-tenancy & security

- **Tenant isolation:** every row is scoped by tenant_id with Postgres RLS forced at the database level.
- **Authentication:** JWT (with per-token JTI for revocation), bcrypt password hashing, login rate-limiting and account lockout, password reset tokens.
- **API access:** scoped API keys for machine clients, plus role-based guards for interactive users.
- **Sectors:** tenants carry a sector (banking, telecom, transport, etc.) driving sector-specific fields and defaults.

4. Module catalogue

Module	Capabilities
Core CRM	Contacts, companies, deals & pipelines (with history), activities/tasks, dashboards.
Sales & Invoicing	Invoices, billing contacts, payments (incl. partial-payment tracking), sales reports, invoice templates & a drag-drop builder, sales settings.
Ticketing	Support tickets, queues, SLA tracking & escalation, CSAT surveys, ticket analytics, complaint handling.
Voice & Voice Bot	Inbound/outbound call logging, recordings and voice analytics (LiveKit); an AI voice bot with IVR config, bot calls and bot-raised tickets.
Email	Shared team inbox with threading, compose/reply and assignment.
Team Messaging	Internal team messaging.
Integrations	SMS gateways, webhooks (queued delivery), Zapier/Make connectors and API bridges.
Analytics	Cross-module reports and dashboards backed by materialized views.
Org & Opportunities	Departments, line-manager hierarchy and opportunity tracking.

5. Super Admin (platform) console

- Tenant lifecycle: create / edit / suspend / reactivate / delete workspaces.
- Plan & module licensing: assign plans and the licensed module + feature set per tenant.
- Platform billing: generate and manage platform invoices and record payments against tenants.
- Platform roles (sub-admins) with their own permission matrix; tenant role management; entitlement sync across tenants.
- Operational: tenant backup/storage tracking, platform metrics and growth charts.

6. Access-control model

Access is enforced through complementary layers, evaluated on every protected request:

Layer	Question	Mechanism
Entitlement	Is the workspace licensed for this feature?	<code>tenants.entitled_features / active_modules</code> → <code>requireEntitlement</code>
Role tier	Does the user's role level allow it?	<code>viewer < agent < manager < tenant_admin</code> → <code>requireScope</code>
Granular permission	Does the role grant this exact action?	per-action booleans (e.g. <code>invoices:edit</code>) → <code>requirePermission</code>

Role types: four built-in system roles (Admin, Manager, Agent, Viewer) seeded per tenant, plus unlimited custom roles. The entitlement is the ceiling — a tenant admin can only grant actions on features the tenant actually licensed.

7. Customer onboarding

- Super admin creates the workspace, choosing the licensed modules and feature-areas agreed with the customer.
- The tenant administrator account is provisioned in the same step, with a one-time temporary password.
- The four standard roles are auto-seeded with sensible defaults; the tenant admin then tailors roles and adds users within the licensed boundary.

8. Data model evolution (migrations)

The schema has evolved across 23 migrations; key milestones:

Phase	Migrations	Focus
Foundation	001–002	Core CRM, multi-tenancy, users, billing
Service modules	003–008	Ticketing, email, voice bot, password reset, complaints, sales invoicing
Access & org	009–015	Forced RLS, roles/sector permissions, milestone templates & routing, threading, line managers, platform roles, permission normalisation
Platform & ops	016–022	Platform invoices, invoice templates, departments & opportunities, webhook queue, analytics materialized views, backups, team messaging
Licensing	023	Tenant module/feature entitlements

9. Current status

Capability area	Status
Core CRM (contacts, companies, deals, activities)	Operational
Sales & invoicing (with partial payments & templates)	Operational
Ticketing, SLA & CSAT	Operational
Voice & AI voice bot	Operational
Email, messaging, integrations, analytics	Operational
Super Admin console & platform billing	Operational
Module/feature licensing + layered enforcement	Operational
Email delivery of credentials + password reset/regenerate	Planned (next)

10. Roadmap / next

- Email the tenant-admin temporary password on creation; add a password reset / regenerate flow.

- Extend granular permission enforcement symmetrically across all modules (currently tightened for Sales).
- Broaden nav-level feature gating coverage and reconcile the migration runner state.